

Nandakumar Vuppalapati

AI Data Engineer — Python, AWS, Databricks, dbt, Spark

Tampa, FL • 424-366-6034 • nandakumaruuppalapati@gmail.com • linkedin.com/in/v-nandakumar

PROFESSIONAL SUMMARY

Data Engineer with progressive experience building and supporting production ETL/ELT pipelines, distributed data processing, and cloud warehouse/lakehouse platforms across financial services and healthcare. Hands-on across Python, SQL, Spark/PySpark, Databricks, dbt, Airflow, Snowflake, AWS, and Azure, with experience spanning ingestion, dimensional modeling, testing, observability, governance, and production support. Recent work extends into ML feature pipelines, MLflow-supported workflows, and LLM/RAG tooling for regulated data and document use cases. Comfortable owning data workflows end to end — from ingestion and transformation through validation, monitoring, troubleshooting, and downstream delivery.

TECHNICAL SKILLS

Languages: Python (Pandas, PySpark), SQL (window functions, CTEs, query tuning), T-SQL, Bash

Cloud: AWS (S3, Glue, Lambda, Step Functions, Redshift, Athena), Azure (Databricks, Data Factory, Synapse Analytics, Blob Storage/ADLS Gen2, Azure SQL)

Data Processing: Apache Spark, PySpark, Databricks, Spark Structured Streaming, Apache Kafka

Orchestration & Transformation: Apache Airflow (DAGs, scheduling, sensors, retries, failure handling), dbt, Azure Data Factory

Warehouse / Lakehouse: Snowflake, Databricks, Delta Lake, AWS Redshift, Azure Synapse; Medallion Architecture (Bronze/Silver/Gold)

Databases: Oracle, PostgreSQL, MySQL, Azure SQL, Databricks SQL, Pinecone (vector database)

DevOps / IaC: Docker, Terraform, GitHub Actions (CI/CD), Git

Data Reliability & Governance: Great Expectations, dbt tests, data contracts, data lineage, PII masking, RBAC, SOX-aware controls, compliance documentation

AI / ML Data Systems: MLflow, feature engineering, XGBoost, RAG, vector retrieval, LangChain, OpenAI API, Pinecone, Feast, Ragas, FastAPI

WORK EXPERIENCE

AI Data Engineer • *Workiva, Remote, USA*

Dec 2025 – Present

- Built and maintained Python/Airflow financial data pipelines across AWS S3, Glue, Lambda, Step Functions, and Redshift, processing 1M+ daily records across payments, risk, settlement, compliance, and downstream analytics while handling production failures, reruns, and recovery validation.
- Developed Snowflake/dbt transformation layers using reusable SQL models, macros, automated tests, and medallion-style data organization, maintaining 30+ financial models that published governed datasets for reporting, risk analysis, and analytical consumers.
- Built Spark/Databricks feature and embedding pipelines over curated financial data for downstream fraud and ML workflows, supporting feature preparation and model lifecycle processes with MLflow.
- Implemented Prometheus/Grafana monitoring across critical ETL workflows, tracing failures through orchestration, transformation, and downstream loading stages and validating data consistency before pipelines returned to normal schedules.
- Built a separate regulatory-document RAG proof of concept with LangChain and OpenAI APIs, combining chunking, embeddings, and retrieval so compliance users could search regulatory content in natural language.

Data Engineer • *Wellmark BCBS, Des Moines, USA*

Aug 2024 – May 2025

- Built Azure Data Factory ingestion pipelines to consolidate 10M+ records from 30+ fragmented legacy healthcare sources through Blob Storage and Azure SQL, standardizing source-to-target processing and applying PII handling controls before downstream transformation.
- Developed PySpark transformation jobs in Azure Databricks to clean, standardize, join, and curate data across medallion-style layers for reporting and analytics, replacing portions of legacy SQL-heavy processing with distributed transformations.
- Created 15+ SQL/dbt transformation models with dbt schema tests and Great Expectations checkpoints, catching schema, null, duplicate, and integrity issues before curated data reached downstream consumers.
- Maintained metadata and lineage documentation and supported repeatable platform delivery with Terraform, Docker, and GitHub Actions, keeping data changes traceable across development and deployment.

- Built an LLM-assisted search prototype across a 1,000+ document vendor-contract repository using document extraction, embeddings, vector retrieval, and natural-language Q&A; evaluated MCP as a possible pattern for broader enterprise document access.

Junior Data Engineer · Sigmoid, Bengaluru, India

Jun 2022 – Dec 2023

- Refactored T-SQL stored procedures and indexing strategies behind recurring finance-reporting workloads, resolving long-running queries and recurring month-end reporting bottlenecks.
- Built a Python-based validation framework with pytest checks integrated into ETL processing, catching schema and null-value issues before data reached stakeholder reports.
- Redesigned a flat reporting structure into a star-schema dimensional model in Azure Synapse, organizing analytical data around fact and dimension tables for more maintainable finance reporting.
- Automated a manual Excel-based weekly reporting workflow with scheduled Azure Data Factory pipelines sourcing directly from Azure SQL, removing repeated spreadsheet preparation and saving roughly 6 hours of manual work each week.
- Supported scheduled ADF pipelines in production, investigated failed runs, validated outputs after recovery, and managed Blob Storage lifecycle policies for staging data.
- Configured ADF schedule triggers and failure alerts and handled recurring data requests from finance analysts, including reruns and output checks when upstream data arrived late.

Data Scientist · Tech Citi Software, Bengaluru, India

Jul 2021 – Dec 2021

- Migrated historical loan and transaction data from Oracle to PostgreSQL, resolving schema and datatype mismatches and validating migrated records for consistent downstream reporting and analysis.
- Built Python and Apache Airflow ETL workflows to ingest 50,000+ daily financial transaction records from CSV and REST API sources into PostgreSQL on a repeatable schedule, replacing manual data-loading steps for downstream reporting and modeling.
- Engineered transaction-level behavioral features and trained an offline XGBoost fraud-classification prototype with SMOTE to address class imbalance, reaching an AUC-ROC of 0.85 in historical-data evaluation.

PROJECTS

Real-Time ML Feature Platform for Fraud Scoring

Kafka, Flink, Feast, Databricks, MLflow, FastAPI, Docker, AWS

- Built to close the gap between batch fraud pipelines and low-latency scoring: Kafka handles event ingestion, Flink computes streaming features, Feast keeps online and offline features aligned, and MLflow/FastAPI support model versioning and serving. The design focuses on feature freshness and training-serving consistency.
- Kept feature computation, feature storage, and serving as separate concerns so the pipeline can evolve without tightly coupling the streaming and API layers; containerized the services for repeatable testing.

CDC Lakehouse with Data Contracts

Debezium, Kafka, Apache Iceberg, dbt Core, Great Expectations, PostgreSQL, Docker

- Built to catch breaking changes close to the source: Debezium captures PostgreSQL row-level changes, Kafka transports them into Iceberg, and dbt plus Great Expectations handle transformation and validation. A schema-contract layer fails incompatible changes before downstream consumers are affected.
- Designed contract checks around schema compatibility and required fields so breaking source changes fail visibly instead of silently propagating into downstream models; containerized the stack to reproduce CDC and recovery scenarios.

RAG Evaluation & Observability Harness

LangChain, OpenAI API, Pinecone, Ragas, FastAPI, Docker, Airflow

- Built around a practical production question — how do you know when a RAG system is getting worse? A golden-question dataset evaluates retrieval and generation separately, with Ragas checks and scheduled Airflow runs surfacing retrieval failures and hallucination regressions.
- Tracked retrieval and generation separately so a weak answer can be traced to missing context versus generation behavior, making the harness useful for regression testing instead of relying on a single aggregate score.

EDUCATION

Master of Computer and Information Sciences — University of South Dakota, Vermillion, SD, USA

2025

Bachelor of Engineering — Indian Institute of Information Technology (IIIT), India

2023

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services

2025